

"Electrocardiography (ECG):



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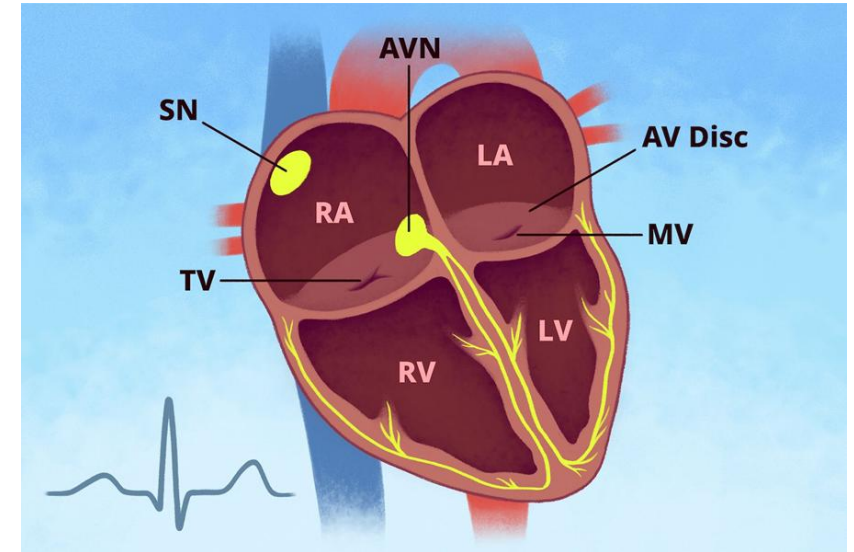
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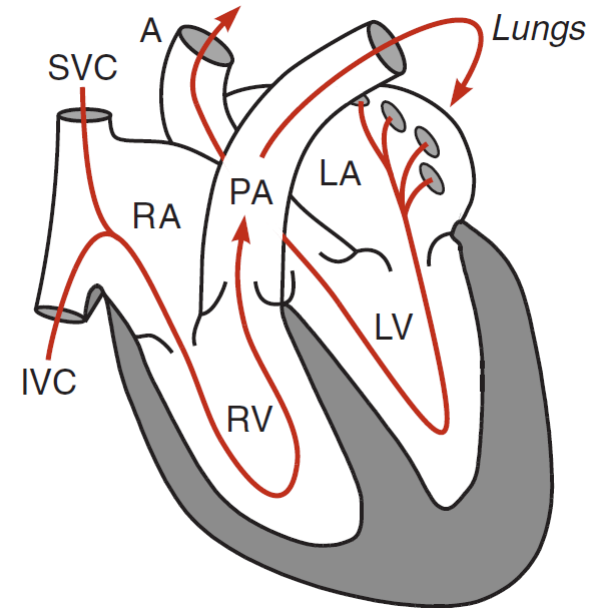
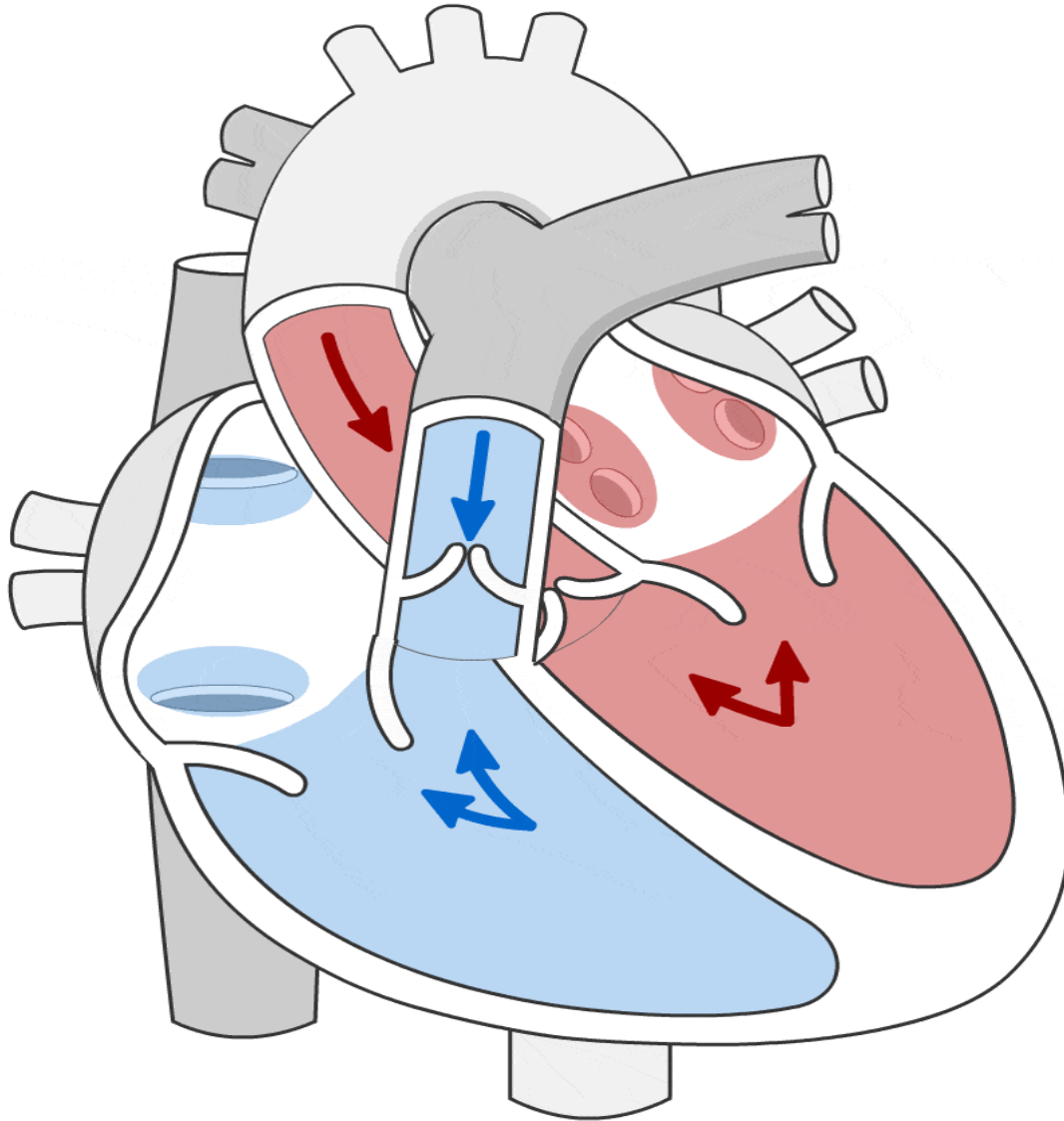
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Electrocardiogram (ECG)

- ❑ ECG; defined as recording the electrical potential which is produced during polarization and depolarization of the cardiomyocyte.
- ❑ The action potentials produced in the heart result in measurable voltages at the skin which can be monitored by external electrodes.
- ❑ These are the largest action potentials measured on the body, producing voltage on the order of 1mV between ECG leads, and it's a direct measurement of voltages produced by the body.



Cardiac Cycle; mechanical events



The Cardiac Conduction System

1

SA Node

The SA node, located in the right atrium, acts as the heart's natural pacemaker, initiating electrical impulses.

2

AV Node

The AV node, located between the atria and ventricles, delays the signal to allow for atrial contraction, before sending it to the ventricles.

3

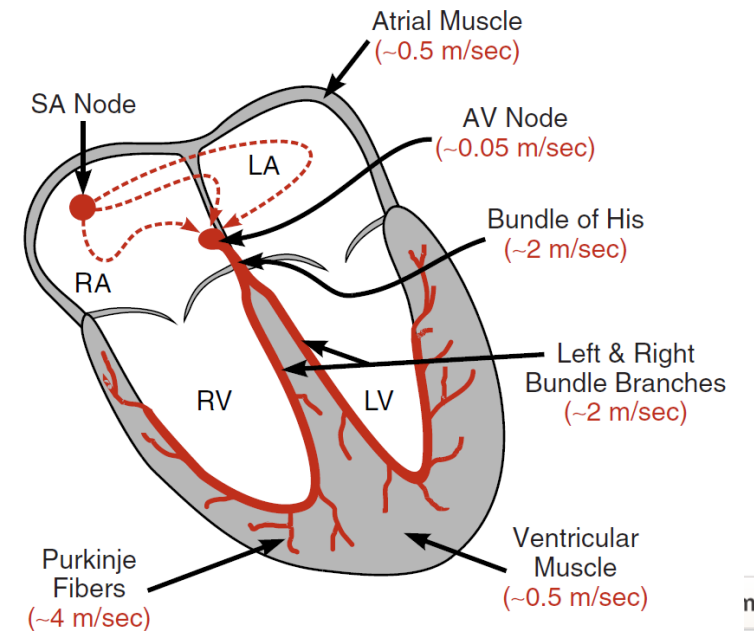
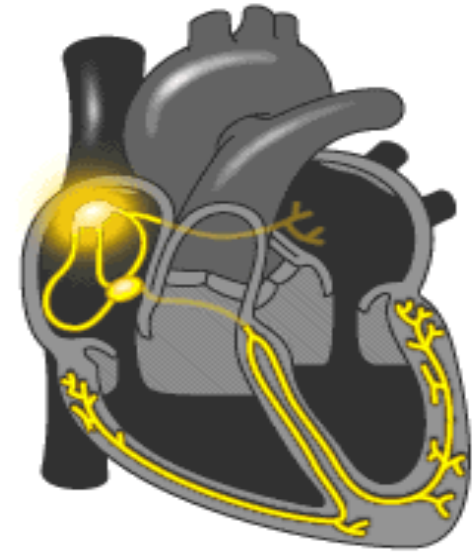
Bundle of His

The Bundle of His, a specialized pathway, conducts the electrical signal from the AV node to the ventricles.

4

Purkinje Fibers

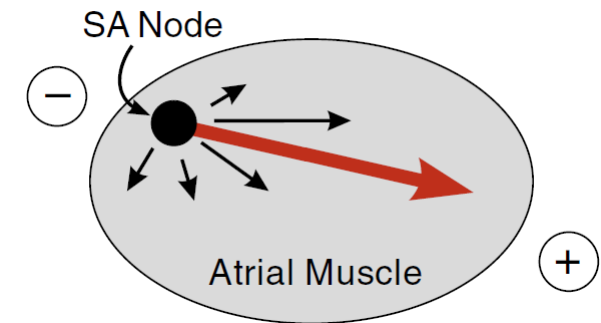
Purkinje fibers, a network of fibers within the ventricles, quickly transmit the electrical signal to the ventricular muscle, ensuring coordinated contraction.



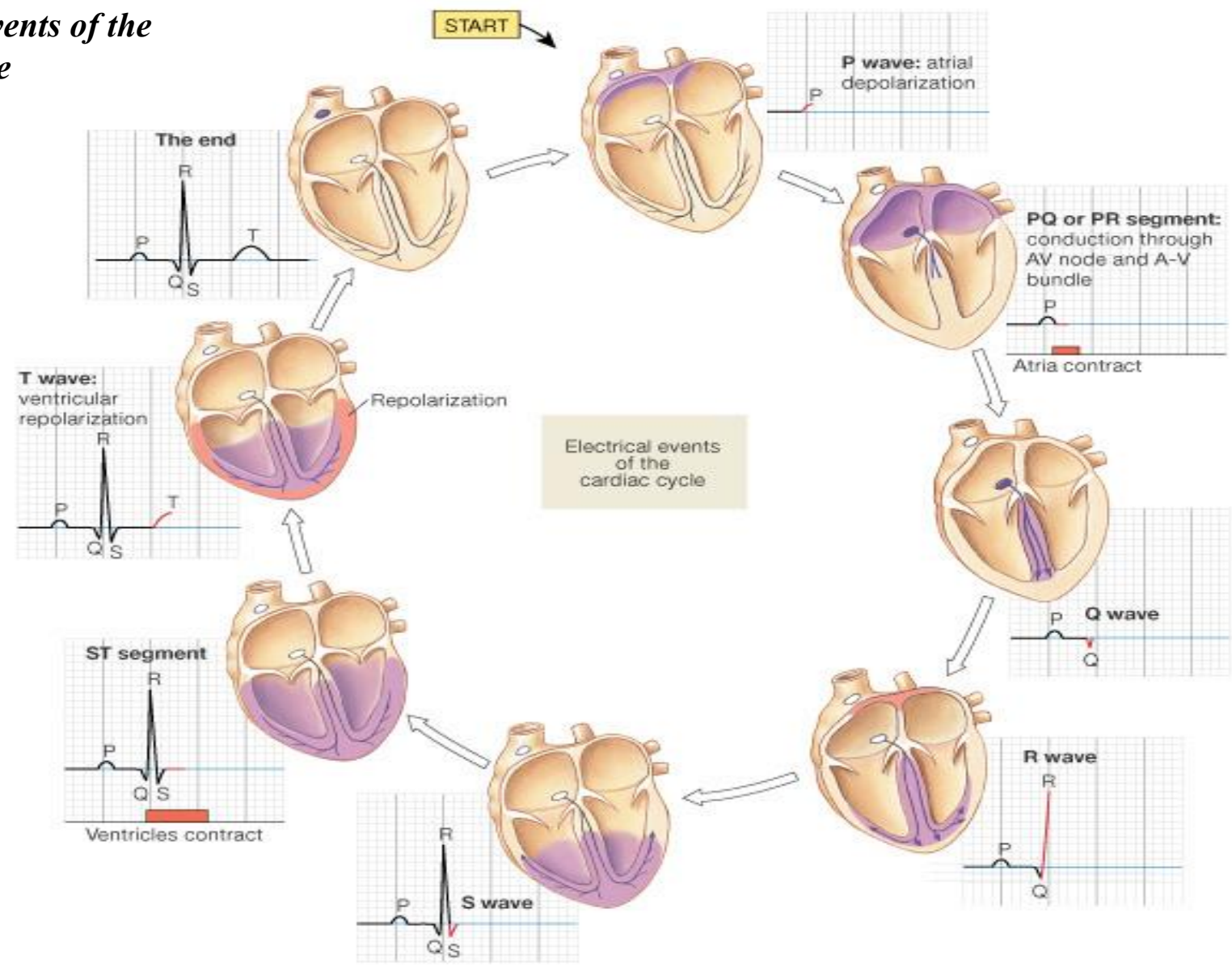
❑ The electrical conducting system of the heart are auto- rhythmic. They will contract regularly without any external stimulation :

❑ Each of these areas of cells could act as a pacemaker. If the SA node is not working, the AV node will act as the pacemaker, and if AV node is also not working, the His- purkinje cells will act as pacemaker.

- **SA Node:** (70-80 bpm).
- **AV Node:** (40-50 bpm).
- **Bundle of His:** Conducts the signal from the AV node to the ventricles.
- **Purkinje Fibers:** (30-40 bpm).



Electrical events of the cardiac cycle



The major electrical events of the normal heart cycle are:

P-wave

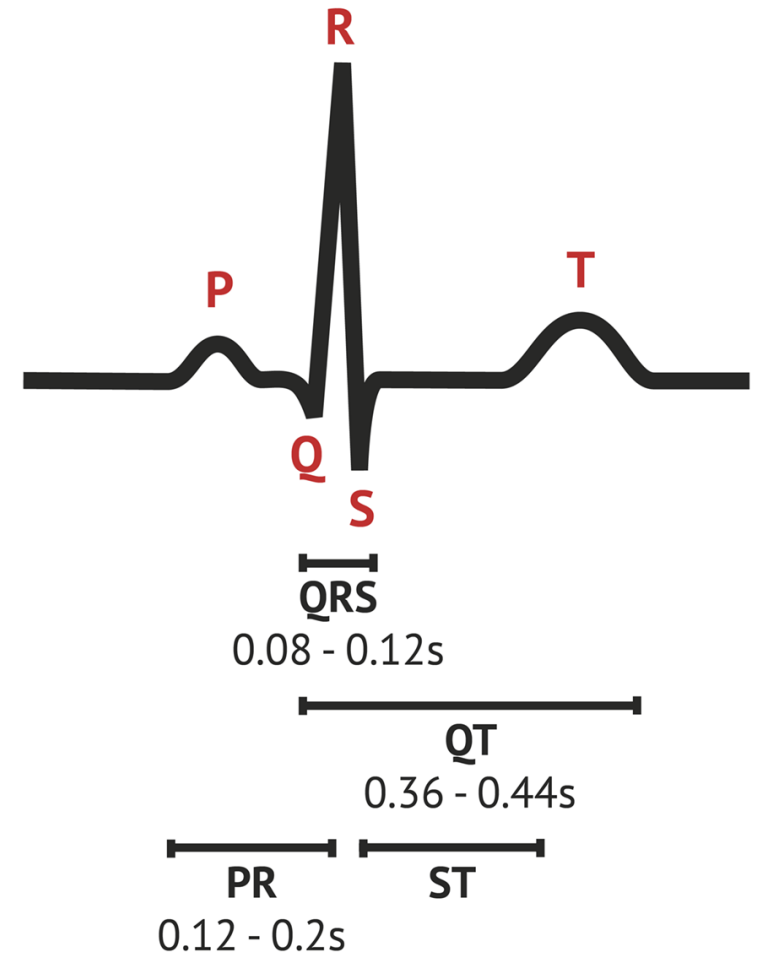
Represents atrial depolarization, indicating electrical activity spreading through the atria.

QRS Complex

Represents ventricular depolarization, a larger complex, indicating electrical activity spreading through the ventricles.

T-wave

Represents ventricular repolarization, the return of ventricular muscle to a resting state after contraction.



ECG basics

Amplitude: 1-5 mV

Bandwidth: 0.05-100 Hz

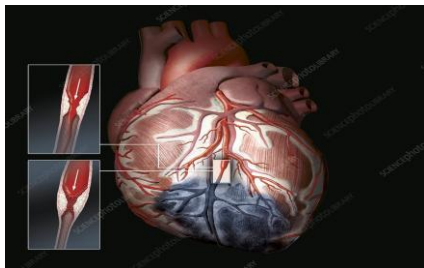
Largest measurement error sources:

- Motion artifacts

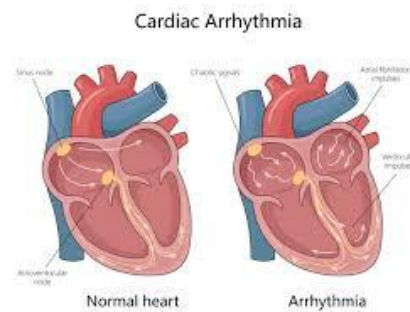
- 50/60 Hz power line interference

Typical applications:

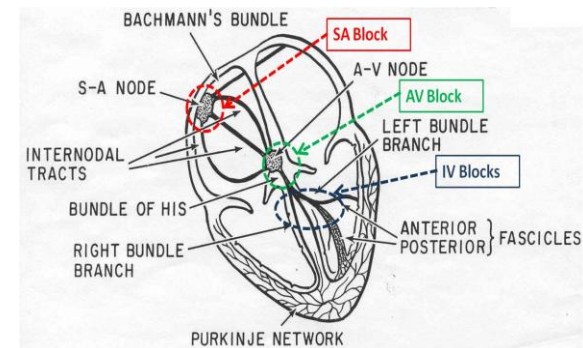
Diagnosis of ischemia



Arrhythmia



Conduction defects



Clinical Applications of ECG

1 Diagnosis of Cardiac Conditions

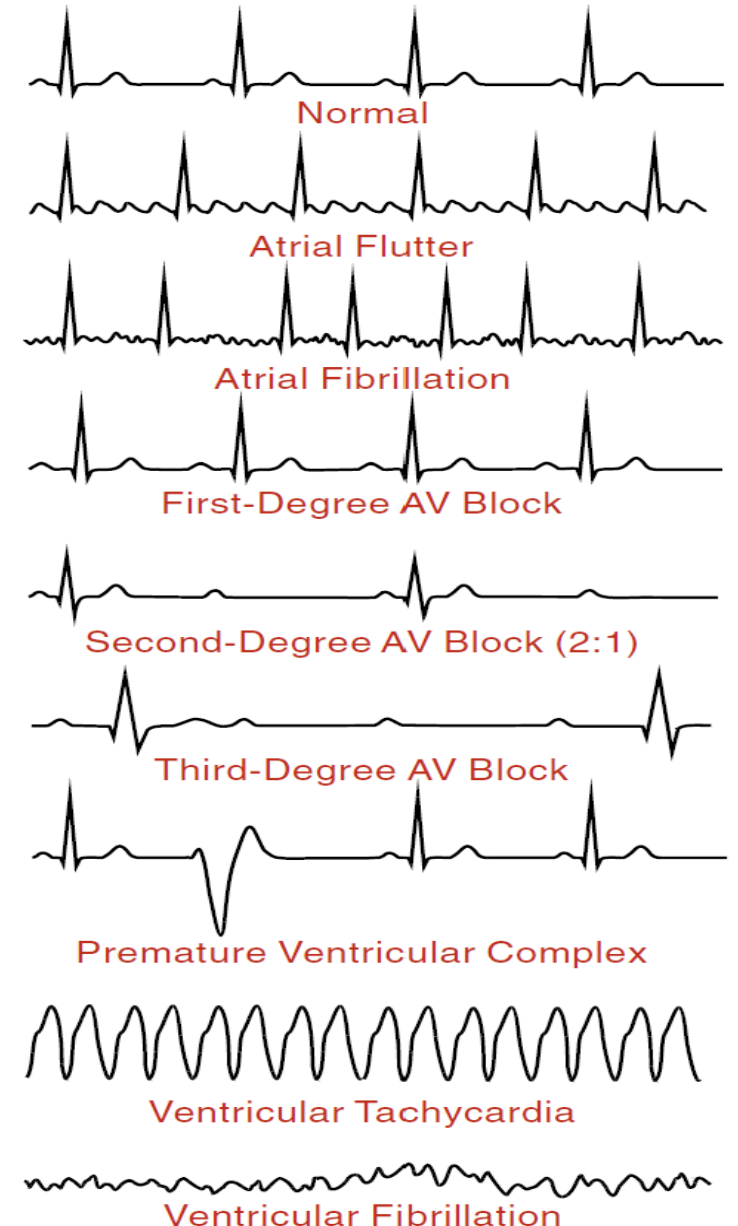
ECG is a valuable tool for diagnosing a wide range of cardiac conditions, myocardial infarction (e.g., atrial fibrillation, ventricular tachycardia).

3 Treatment Evaluation

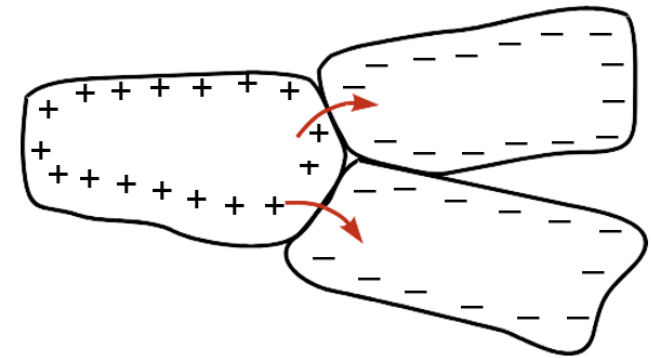
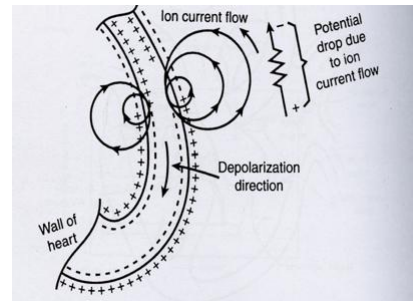
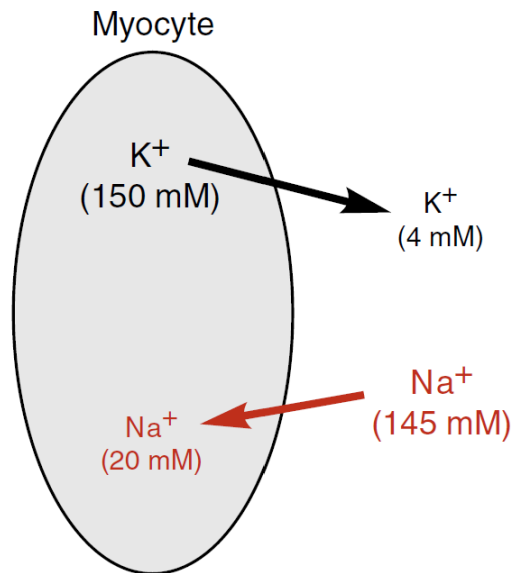
ECG helps evaluate the effectiveness of treatments for cardiac conditions and monitor patient progress over time.

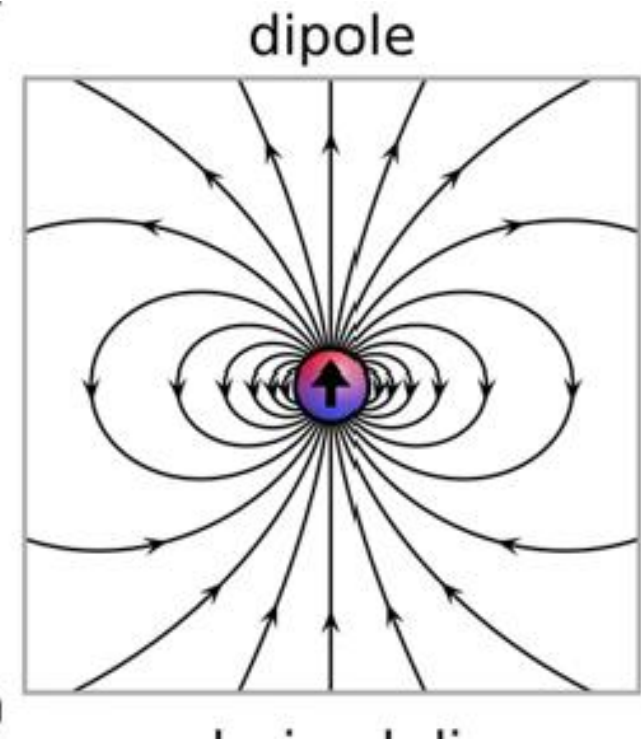
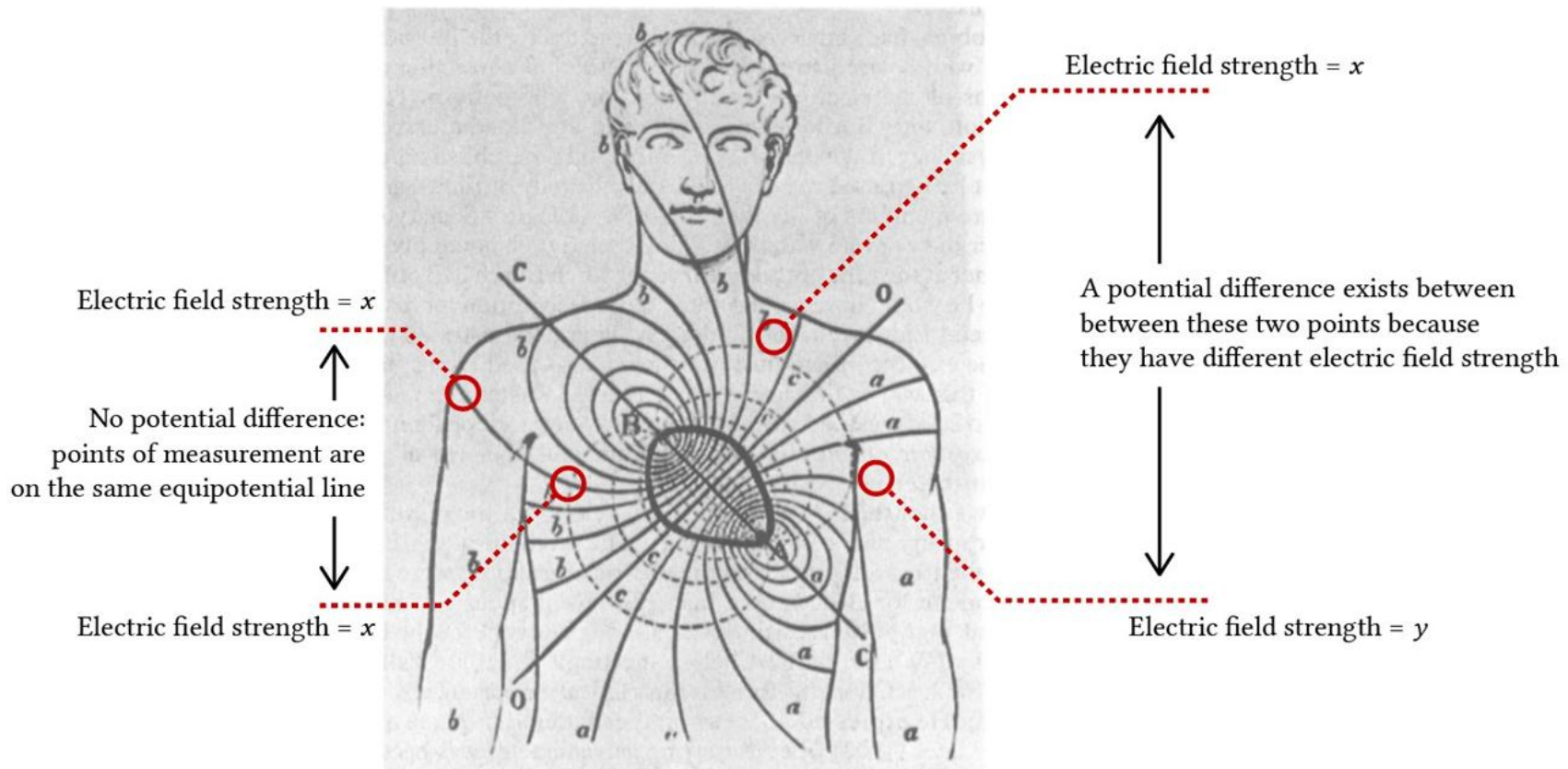
2 Monitoring Heart Function

ECG can monitor heart function during procedures, surgery and exercise stress to assess heart response to stress.

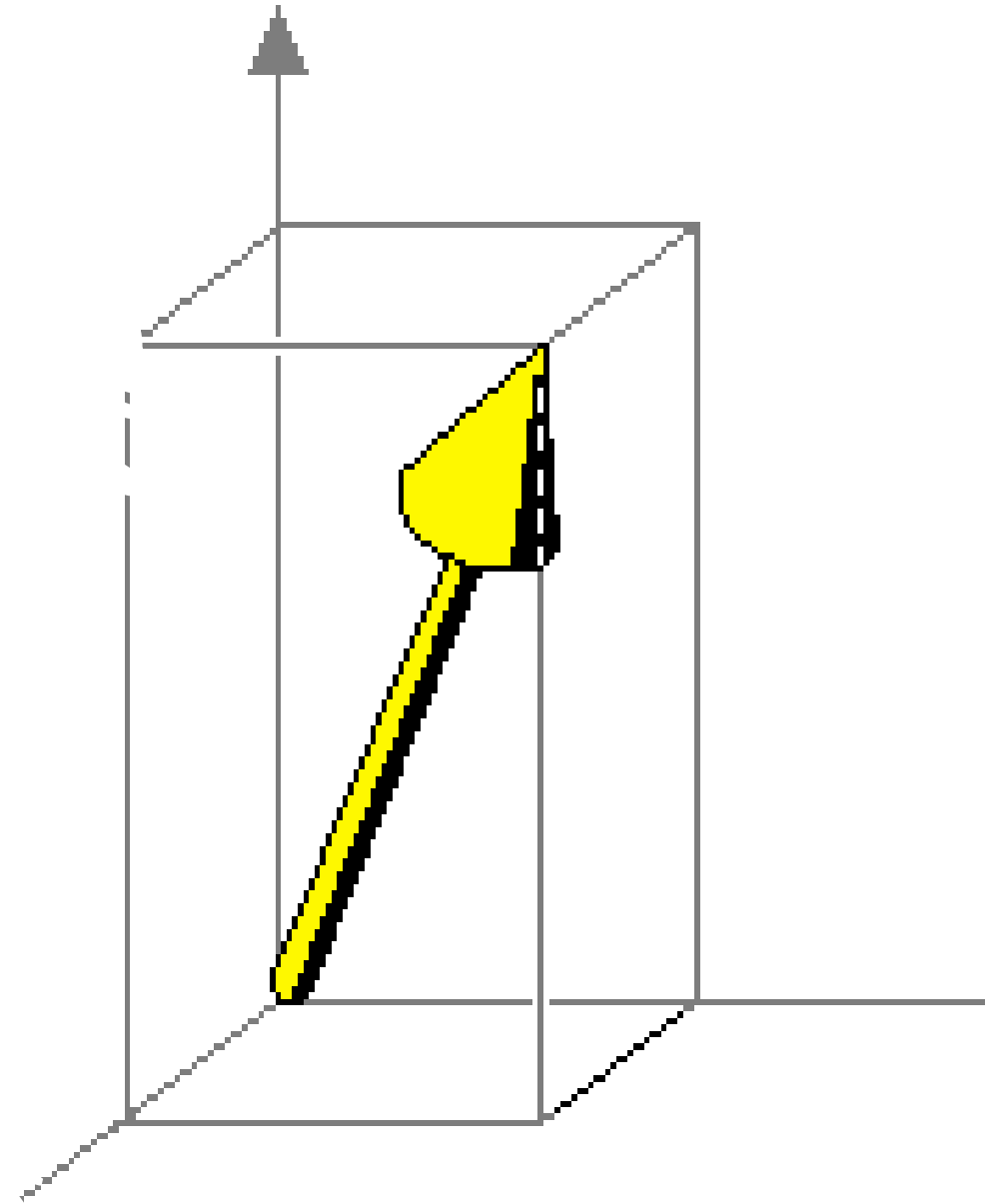


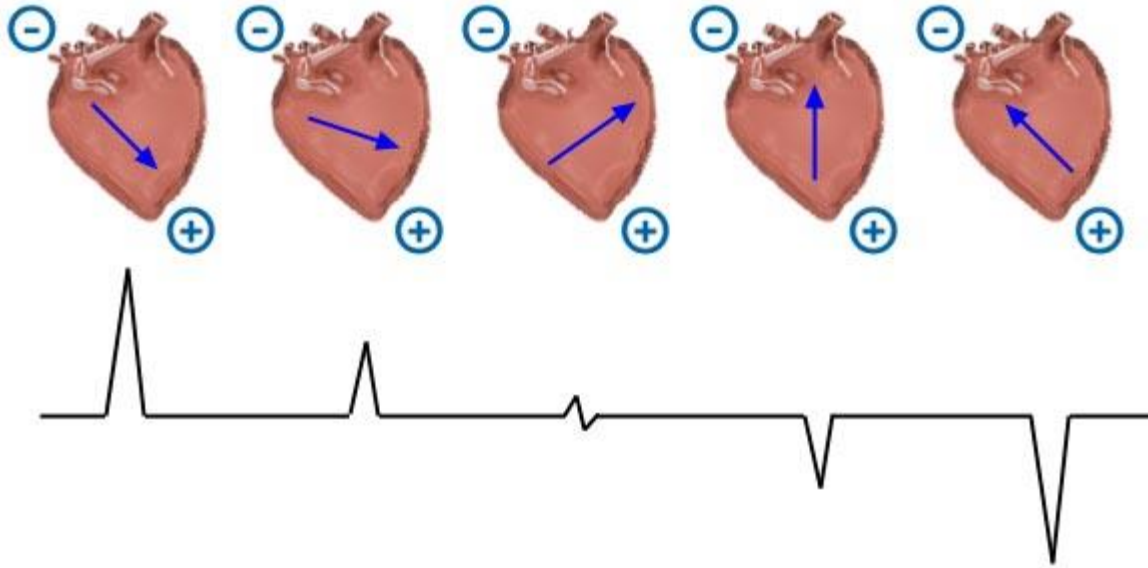
- ❑ Nerve and muscle of the heart can be regarded as sources of electricity enclosed in an electrical conductor.





- The electrical (cardiac) potential that we measure on the body's surface is
 - The Instantaneous projection of the electric dipole vector in a particular direction.
 - As the vector changes with time, so does the projection

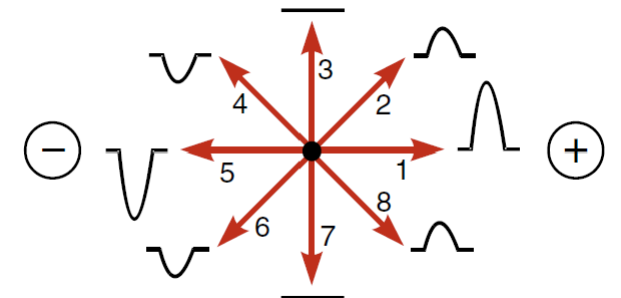
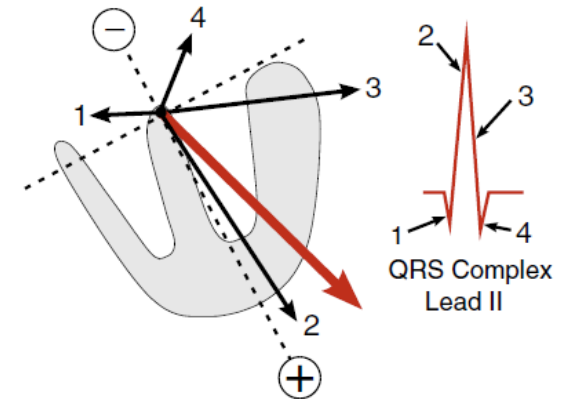
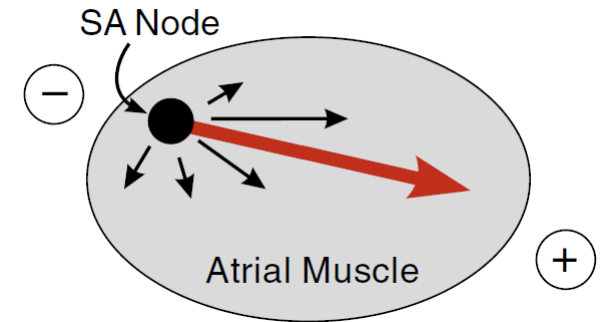




If the dipole vector directed towards the positive electrode of lead – ECG will show positive wave

If the dipole vector directed toward the negative electrode lead – electrocardiogram will show negative wave

If the dipole vector is perpendicular to the axis of abduction, then electrocardiogram writes isoline.

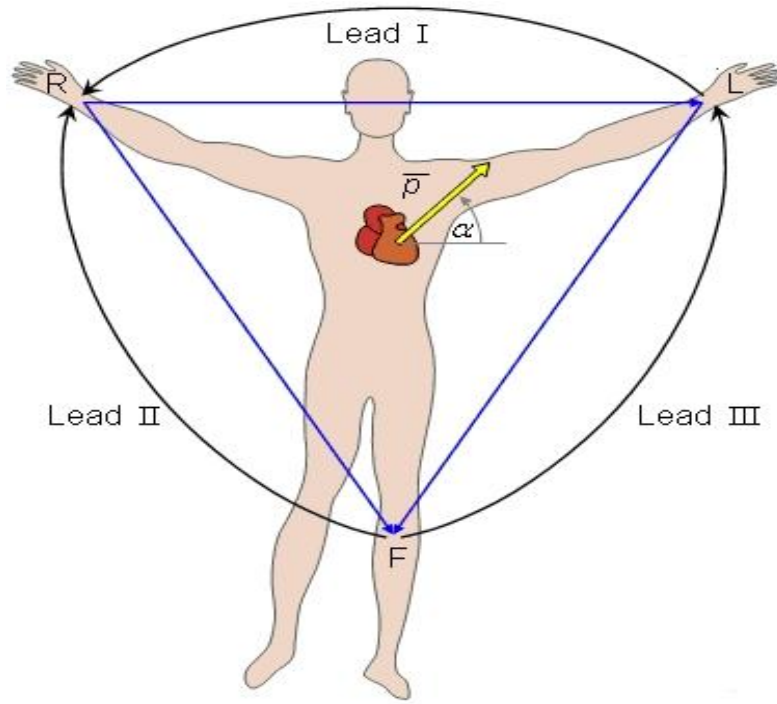


12-Lead ECG measurement

Proper electrode placement is crucial for obtaining accurate electrocardiogram (ECG) readings. A standard 12-lead ECG uses 9 electrodes: 3 limb electrodes and six precordial (chest) electrodes.

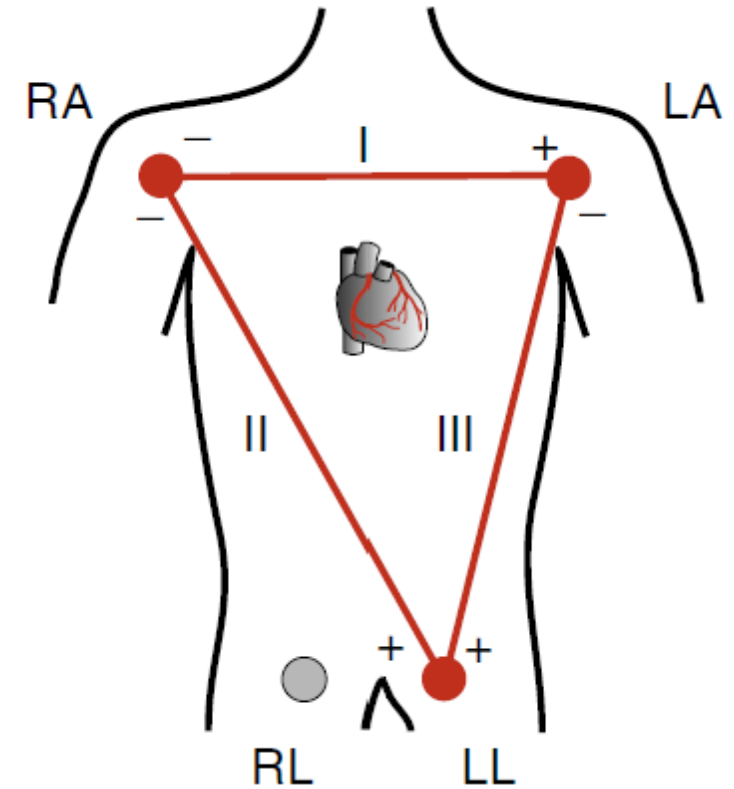
- Most widely used ECG measurement setup in clinical environment
- Signal is measured non-invasively with 9 electrodes
- Lots of measurement data and international reference databases
- Well-known measurement and diagnosis practices

- Einthoven leads: I, II & III

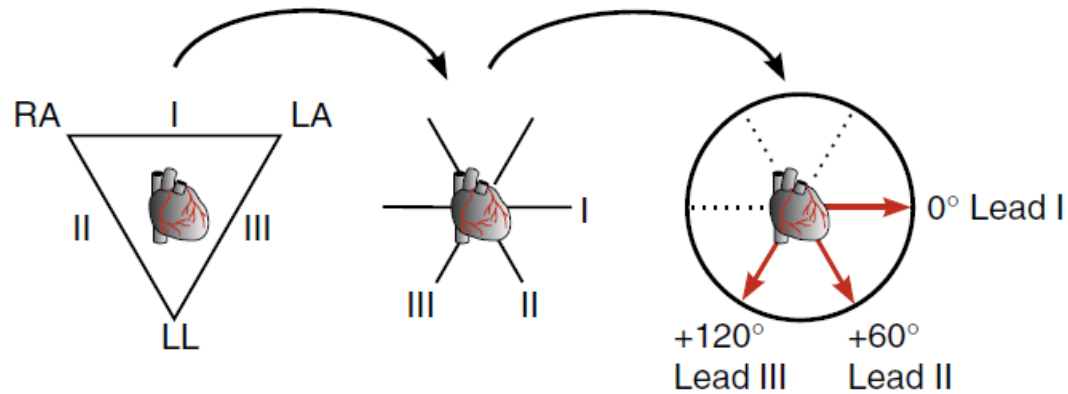
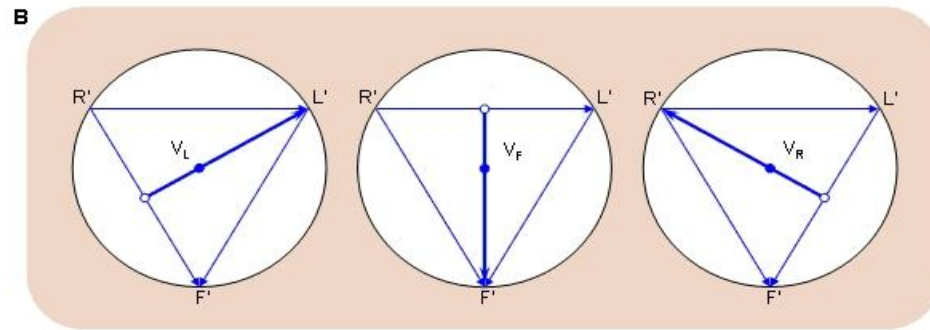
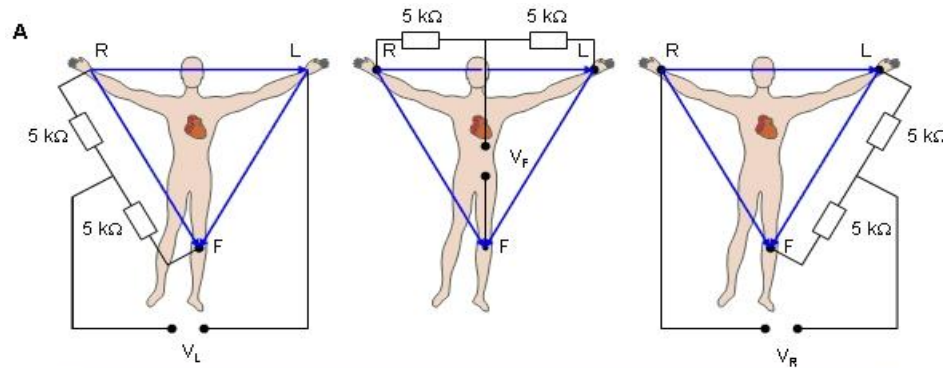


Limb Electrode Placement:

- **Right Arm (RA):** Placed on the right forearm or wrist.
- **Left Arm (LA):** Placed on the left forearm or wrist.
- **Right Leg (RL):** Placed on the right lower leg or ankle.
- **Left Leg (LL):** Placed on the left lower leg or ankle.

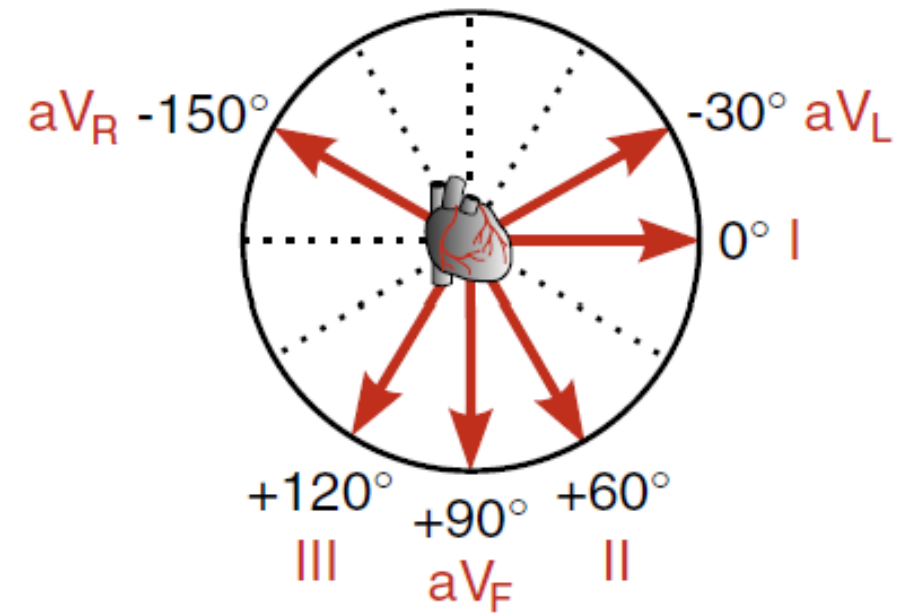


Goldberger augmented leads: V_R , V_L & V_F



Einthoven's Triangle

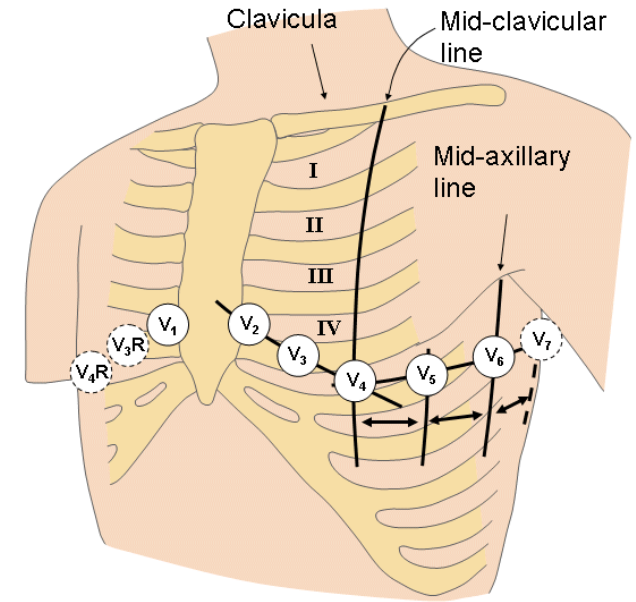
Axial Reference System



Transverse plane leads: V_1 - V_6

Precordial (Chest) Electrode Placement:

- V1**: Fourth intercostal space at the right sternal border.
- V2**: Fourth intercostal space at the left sternal border.
- V4**: Fifth intercostal space at the midclavicular line.
- V3**: Midway between V2 and V4.
- V5**: Level with V4 at the anterior axillary line.
- V6**: Level with V4 at the midaxillary line.

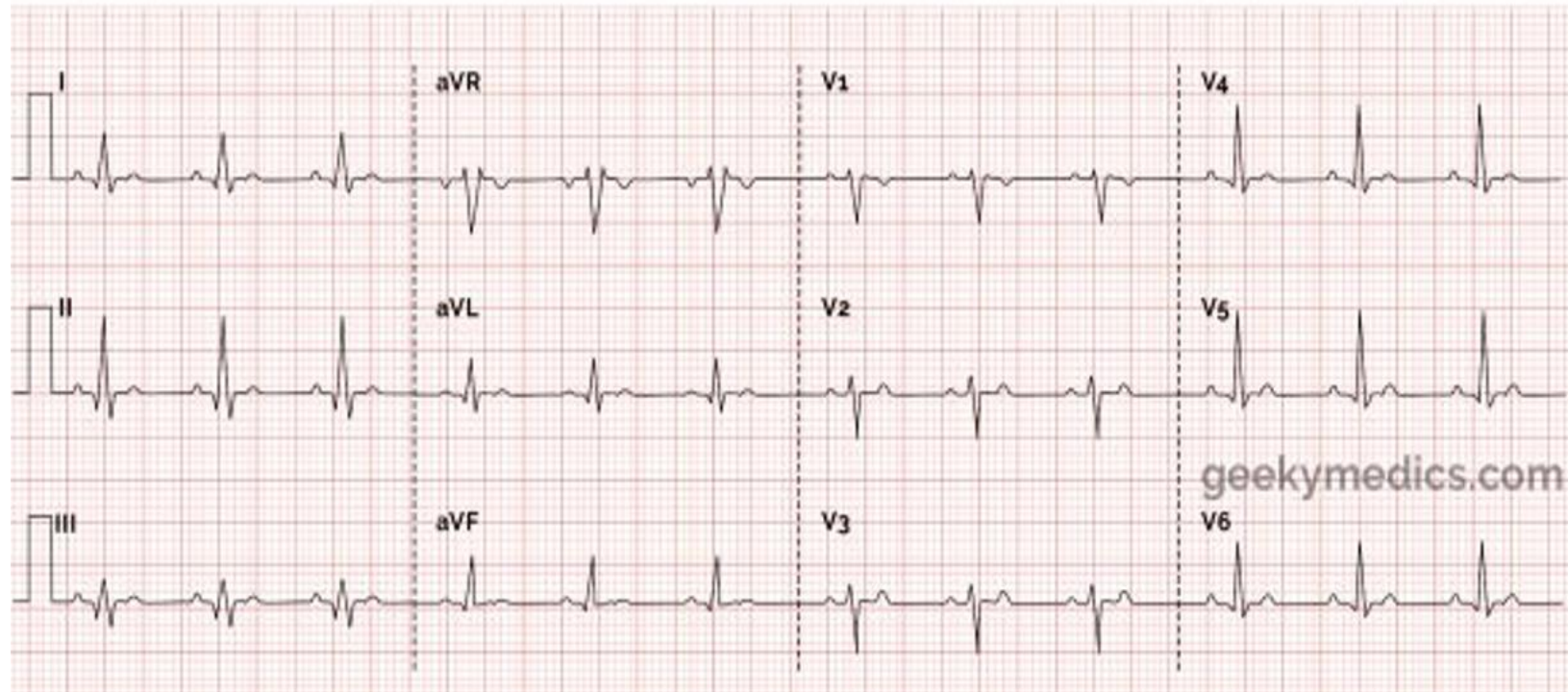


<https://www.youtube.com/watch?v=kWLbSx9BNbU>

Figure shows the 12 ECG signals from a normal heart.

Three of the ECG electrodes are placed on the right and left arms and the left leg, and the voltages across the three pairs of these electrodes are monitored (along with the signals from the other probes).

Normal ECG



■ Standardization

■ Rate

1. RR interval

2. Heart rate

■ Rhythm

3. PP interval

4. P wave

■ width, height, shape, etc.

5. PR interval

6. QRS

■ width (and height)

■ axis

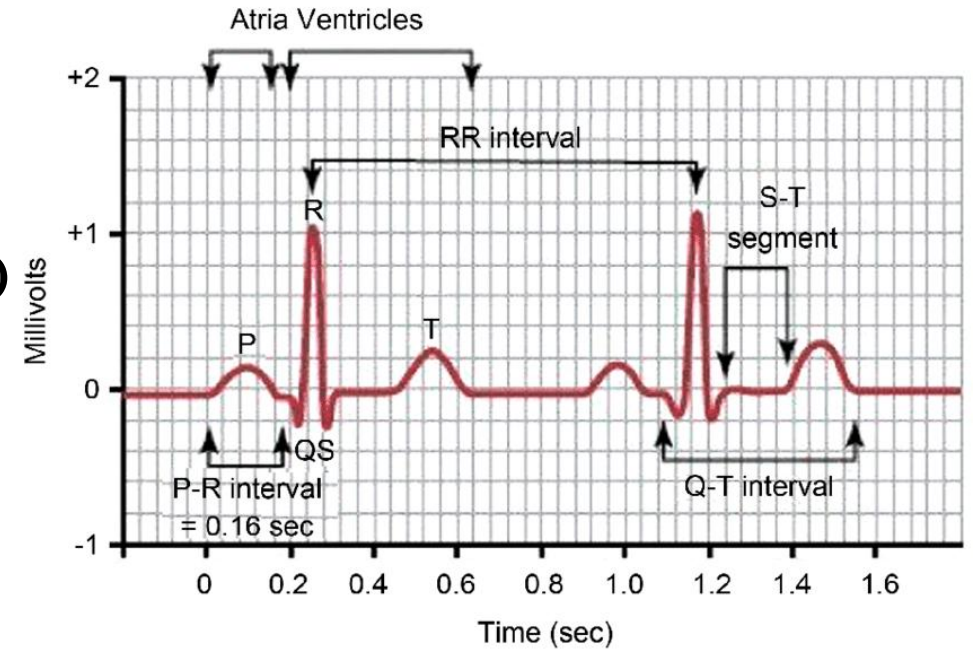
■ R wave
progression

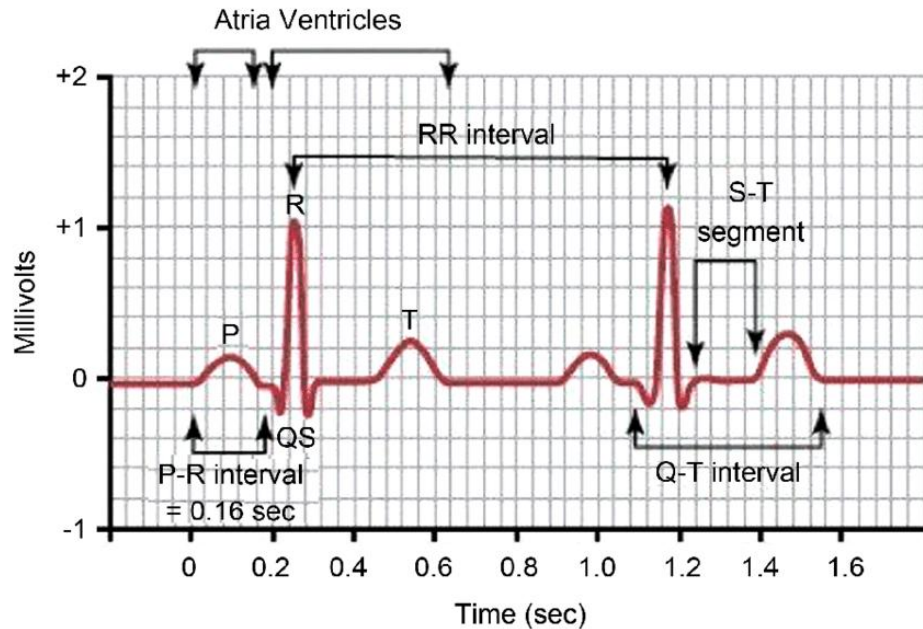
■ abnormal Q waves

■ ST segment

■ T waves

■ QT interval





1. Normal vs. Abnormal Examples:

1. A comparison of a **normal sinus rhythm** with common abnormalities, such as:

1. ST elevation (indicating myocardial infarction).

2. Arrhythmias (e.g., atrial fibrillation, ventricular tachycardia).

3. Heart block (e.g., prolonged PR interval).

A 58-year-old male with a history of hypertension and smoking presents to the emergency department with complaints of severe chest pain that started 2 hours ago. The pain is described as a crushing sensation in the center of his chest, radiating to his left arm and jaw. He also reports associated symptoms of shortness of breath, nausea, and diaphoresis (excessive sweating).

Which of the following is the single most appropriate immediate investigation to confirm the diagnosis?

- A. Chest X-ray
- B. Echocardiography
- C. Electrocardiogram (ECG)
- D. Exercise stress test
- E. CT coronary angiography



Limitations and Emerging Technologies in ECG

1

Limitations

ECG can miss some heart conditions, may be influenced by external factors, and requires proper electrode placement.

2

Emerging Technologies

Wearable ECG devices, AI-powered analysis, and improved imaging techniques are advancing ECG technology.

- **Wearable ECG devices:** Mention popular devices like Apple Watch or Fitbit that offer ECG monitoring.
- **AI-powered analysis:** Explain how AI is being used to improve ECG interpretation and diagnosis.
- **Improved imaging techniques:** Briefly mention how ECG is being integrated with other imaging modalities like MRI or CT.

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